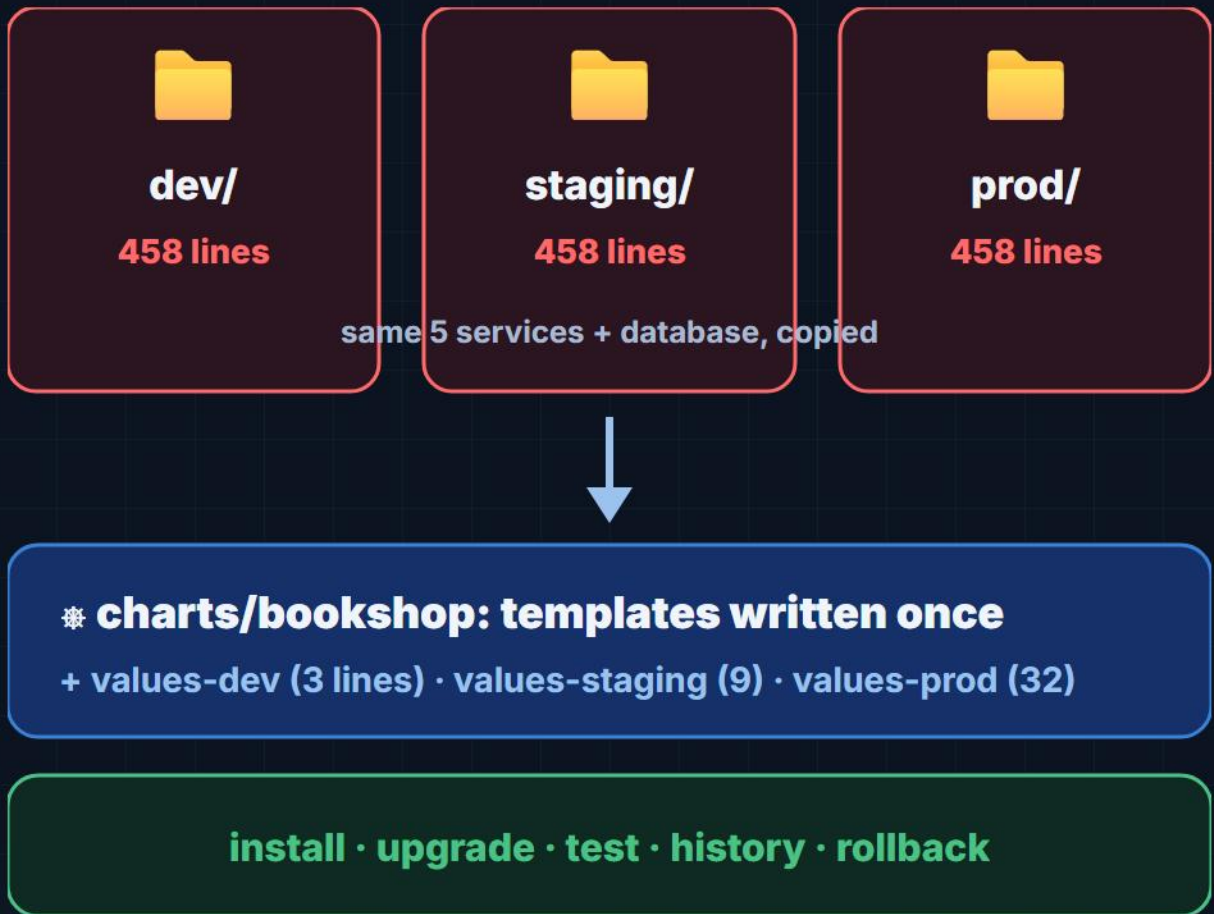


1,374 lines of YAML → **one Helm chart**

The same app in dev, staging and prod, without copying it three times.



THE PAIN

What copied YAML costs you

fine for one environment, painful for three.



1 change = 3 edits

a new image tag lives in three files; forget one, and environments drift



No history

what runs in staging?
what changed on Tuesday?
how do we undo it?



Passwords in Git

so the folder can be applied as-is; base64 is not encryption



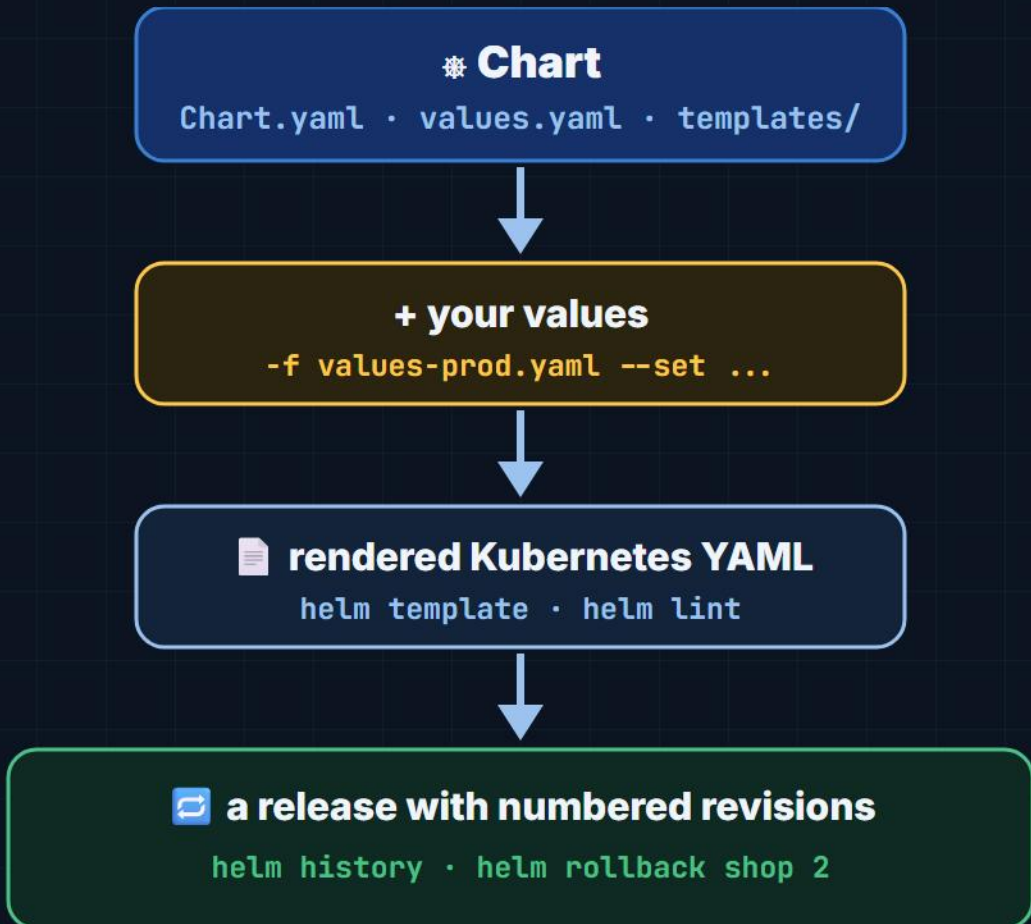
Sharing = copying

another team wants it?
they copy your folder and edit it too

THE IDEA

Package · template · **release**

Helm renders plain YAML before Kubernetes sees anything.



THE DISTINCTION EVERYONE MIXES UP

Chart version $\neq$ app version

two numbers, two meanings, both in every helm list.

version: 1.2.0

the CHART

templates, defaults, options
the package itself

appVersion: "1.0.0"

the APPLICATION

the default image tag
the software inside

new chart option (a PDB)

1.2.0 $\rightarrow$ 1.3.0 $\cdot$ app same

new app release

app 2.0.0 $\rightarrow$ 2.1.0 + chart bump

appVersion: 1.10 (unquoted)

$\rightarrow$ image tag 1.1 

a published chart version never changes contents

VALUES PRECEDENCE

Who wins? The last one.

predict it, then render it: helm template shows the answer.

`values.yaml`

`logLevel: info`

`-f team.yaml`

`logLevel: warn`

`-f values-prod.yaml`

`logLevel: error ✓`

`--set replicaCount=5`

wins over all

maps merge key by key · lists are replaced

`replicacount: 3 (typo)` → silently ignored

`values.schema.json` → rejected at lint time

THE CAPSTONE'S RELEASE HISTORY

A week, in **four lines**

helm history shop, recorded in the lab.

1 **Install complete** **superseded**

2 **node-api 1.1.0 (reviewed diff)** **superseded**

3 **java-api 2.0.0: never published** **failed**

4 **Rollback to 2** **deployed**

the old Pod kept serving during the failure

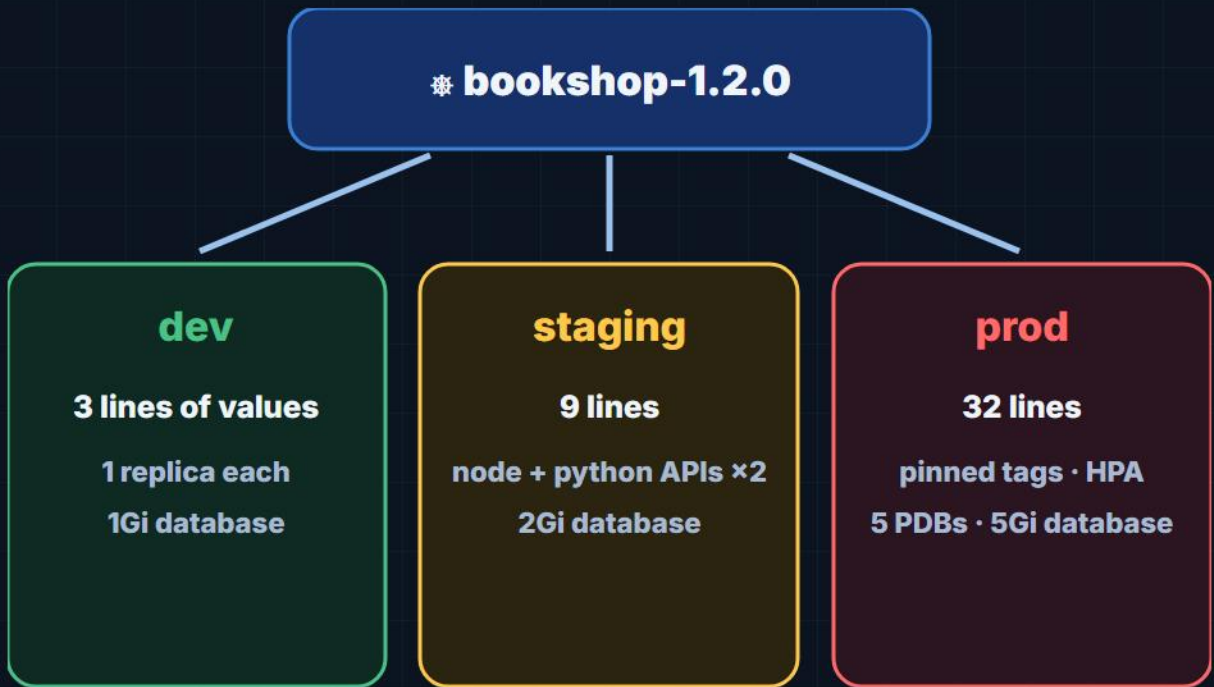
helm test after the rollback: all checks passed

a rollback restores objects and values, not data

MULTIPLE ENVIRONMENTS

One chart version, **three environments**

only values differ; every env runs the templates prod will run.



dev: Succeeded · staging: Succeeded · prod: Succeeded

helm test of each environment: the promotion gate

12 TROUBLESHOOTING SCENARIOS

Errors you will **meet at work**

each: problem → investigation → root cause → fix → verification.

`parse error ... unexpected EOF`

`spec.selector: ... field is immutable`

`another operation ... is in progress`

`conflict with "kubectl-patch"`

`secret "shop-db-prod" not found`

`missing in charts/ directory: postgres`

+ wrong values · bad image · OOMKilled · drift · values conflict

STUDY MATERIAL

Everything to learn it properly

free, in one repository

17

tested labs

12

troubleshooting scenarios

12

challenges

16

concept pages

24

course chapters

3

charts published (OCI)

🔍 19-step capstone · 📄 cheat sheet · triage script

🎬 24-chapter video (00:00) · full + silent version

📄 00-page study guide PDF

💬 71-term glossary · 25 interview questions

RUN IT YOURSELF

Six commands to a tested release

Docker, kind, kubectl, Helm 4. No cloud account.

```
1 $ git clone .../sufyan-devops-helm-from-zero
```

```
2 $ kind create cluster --config ...
```

```
3 $ helm install traefik traefik/traefik ...
```

```
4 $ helm dependency build charts/bookshop
```

```
5 $ helm upgrade --install shop ... --wait
```

```
6 $ helm test shop -n bookshop-dev
```

all checks passed ✓

every lesson shows the real output to compare

FREE AND OPEN SOURCE

Chart it. Ship it. **Roll it back.**

How many copies of the same YAML does your team keep?



The repository

[github.com/sufyanahmadkamboh/
sufyan-devops-helm-from-zero](https://github.com/sufyanahmadkamboh/sufyan-devops-helm-from-zero)



All my projects

sufyanahmadkamboh.github.io

[charts](#) · [values](#) · [releases](#) · [12 failures](#) · [video](#) · [PDF](#)